

EX 01:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO FIND THE ROOTS OF QUADRATIC EQUATION USING ELSE IF LADDER

Source Code:

```
#include <stdio.h>

#include <math.h>

void main()

{

    float a, b, c;

    float discriminant, root1, root2, realPart, imagPart;

    printf("Enter coefficients a, b and c: ");

    scanf("%f %f %f", &a, &b, &c);

    discriminant = b * b - 4 * a * c;

    if (discriminant > 0) // Using else if ladder

    {

        root1 = (-b + sqrt(discriminant)) / (2 * a); // Two distinct real roots

        root2 = (-b - sqrt(discriminant)) / (2 * a);

        printf("Roots are real and different.\n");

        printf("Root 1 = %.2f\n", root1);

        printf("Root 2 = %.2f\n", root2);

    }

    else if (discriminant == 0) {

        // Two equal real roots

        root1 = -b / (2 * a);
```

```

        printf("Roots are real and equal.\n");
        printf("Root = %.2f\n", root1);
    }
    else {
        // Complex roots
        realPart = -b / (2 * a);
        imagPart = sqrt(-discriminant) / (2 * a);
        printf("Roots are complex and imaginary.\n");
        printf("Root 1 = %.2f + %.2fi\n", realPart, imagPart);
        printf("Root 2 = %.2f - %.2fi\n", realPart, imagPart);
    }
}

```

Output:

```

Enter the coefficients a, b and c: 1 -3 2
Roots are real and different
Root 1: 2.00
Root 2: 1.00

```

```

Enter the coefficients a, b and c: 1 -2 1
Roots are real and equal
Root 1: 1.00

```

```

Enter the coefficients a, b and c: 1 2 5
Roots are complex and imaginary
Root 1: -1.00 + 2.00i
Root 2: -1.00 - 2.00i

```

EX 02:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO READ TWO INTEGER VALUES AND AN OPERATOR HAS A CHARACTER AND PERFORM BASIC ARITHMETIC OPERATION ON THEM USING SWITCH CASE (+, -, *, /)

Source Code:

```
#include <stdio.h>

main()
{
    int num1, num2;
    char operator;
    int result;

    printf("Enter first integer: ");
    scanf("%d", &num1);
    printf("Enter second integer: ");
    scanf("%d", &num2);
    printf("Enter an operator (+, -, *, /, %%): ");
    scanf(" %c", &operator); // Notice the space before %c to consume any
    leftover newline

    switch (operator) {
        case '+':
            result = num1 + num2;
            printf("%d %c %d = %d\n", num1, operator, num2, result);
            break;
        case '-':
            result = num1 - num2;
            printf("%d %c %d = %d\n", num1, operator, num2, result);
```

```
        break;
    case '*':
        result = num1 * num2;
        printf("%d %c %d = %d\n", num1, operator, num2, result);
        break;
    case '/':        // Check for division by zero
        if (num2 != 0) {
            result = num1 / num2;
            printf("%d %c %d = %d\n", num1, operator, num2, result);
        } else {
            printf("Error: Division by zero is not allowed.\n");
        }
        break;
    default:
        printf("Error: Invalid operator.\n");
        break;
}
}
```

Output:

```
Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): +
100 + 50 = 150
```

```
Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): -
100 - 50 = 50
```

```
Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): *
100 * 50 = 5000
```

```
Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): /
100 / 50 = 2
```

EX 03:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO REVERSE A NUMBER TO CHECK IT IS A VALID PALINDROME AND TO CALCULATE THE SUM OF DIGITS

Source Code:

```
#include <stdio.h>

main() {

    int num, originalNum, reversedNum = 0, digit, sumOfDigits = 0;

    printf("Enter an integer: ");

    scanf("%d", &num);

    originalNum = num;

    while (num != 0){

        digit = num % 10;        // Get the last digit

        reversedNum = reversedNum * 10 + digit; // Construct the reversed
number

        sumOfDigits += digit;    // Add the digit to the sum

        num /= 10;              // Remove the last digit from num

    }

    if (originalNum == reversedNum){        // Check if the original
number is a palindrome

        printf("%d is a palindrome.\n", originalNum);

    } else{

        printf("%d is not a palindrome.\n", originalNum);

    }

    printf("Reversed number: %d\n", reversedNum); // Print the results

    printf("Sum of digits: %d\n", sumOfDigits);
```

```
}
```

Output:

```
Enter an integer: 123
123 is not a palindrome
Reversed number: 321
Sum of digits: 6
```

```
Enter an integer: 232
232 is a palindrome
Reversed number: 232
Sum of digits: 7
```

EX 04:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO GENERATE N FIBONACCI NUMBER

Source Code:

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int f1=0,f2=1,f3,i,n;

    clrscr();

    printf("Enter the value for N: ");

    scanf("%d",&n);

    for(i=1;i<=n;i++)
    {
        printf("%d\t",f1);

        f3=f1+f2;

        f1=f2;

        f2=f3;
    }

    getch();
}
```

Output:

```
Enter the value for N: 10
0      1      1      2      3      5      8      13     21     34
```


EX 05:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO FIND THE GIVEN NUMBER IS A PRIME NUMBER OR NOT

Source Code:

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int n,i,flag=0;

    clrscr();

    printf("Enter a positive integer : ");

    scanf("%d",&n);

    for(i=2;i<=n/2;++i)

    {

        if(n%i==0)

        {

            flag=1;

            break;

        }

    }

    if(flag==0)

        printf("%d is a prime number",n);

    else

        printf("%d is not a prime number",n);
```

```
        getch();  
    }
```

Output:

```
Enter a positive integer: 2  
2 is a prime number
```

```
Enter a positive integer: 22  
22 is not a prime number
```

EX 06:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO COUNT OCCURANCES OF EACH CHARACTER IN A GIVEN STRING

Source Code:

```
#include<stdio.h>

#include<conio.h>

int main()
{
    char str[20],ch;
    int count=0,i;
    clrscr();
    printf("Enter a string: ");
    scanf("%s",str);
    printf("Enter the character to be searched: ");
    scanf(" %c",&ch);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]==ch)
            count++;
    }
    if(count==0)
        printf("Character '%c' is not present",ch);
    else
        printf("Occurence of character '%c':%d",ch,count);
```

```
    getch();  
}
```

Output:

```
Enter a string: Hello123#  
Enter the character to be searched: l  
Occurence of character 'l':2
```

```
Enter a string: Hello123#  
Enter the character to be searched: a  
Character 'a' is not present
```

EX 07:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO READ WITH ALPHABETS, DIGITS, AND SPECIAL CHARACTERS AND CONVERT LETTERS INTO UPPERCASE AND LOWERCASE AND VICE-VERSA AND RETAIN THE DIGITS AND SPECIAL CHARACTERS AS IT IS

Source Code:

```
#include<stdio.h>

void convertCase(char *str);

main()
{
    char str[100];
    clrscr();
    printf("Enter the String: ");
    gets(str);
    convertCase(str);
    printf("Converted string: %s", str);
    getch();
}

void convertCase(char *str)
{
    while (*str)
    {
        if (islower(*str))
        {
            *str = toupper(*str);
        }
    }
}
```

```
        }  
        else if (isupper(*str))  
        {  
            *str = tolower(*str);  
        }  
        str++;  
    }  
}
```

Output:

```
Enter the String: Sacred Heart College  
Converted String: sACRED hEART cOLLEGE
```

EX 08:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

**PURPOSE: WRITE C PROGRAM TO SEARCH FOR NUMBER OF OCCURANCE OF
NUMBER IN A LIST OF NUMBERS USING ONE DIMENSIONAL ARRAY AND ALSO
DISPLAY ITS POSITION**

Source Code:

```
#include <stdio.h>

void searchOccurrences(int arr[], int, int);

int main(){
    int arr[20];

    int size, i, target;

    clrscr();

    printf("Enter the size of the array: ");

    scanf("%d", &size);

    if (size <= 0 || size > 20) {
        printf("\nInvalid size!\n");
        return 1;
    }

    printf("Enter %d elements: ", size);

    for (i = 0; i < size; i++){
        scanf("%d", &arr[i]);
    }

    printf("Enter the search element: ");

    scanf("%d", &target);

    searchOccurrences(arr, size, target);
}
```

```
    getch();  
    return 0;  
}  
  
void searchOccurrences(int arr[], int size, int target){  
    int count = 0,i;  
    printf("Position(s) of %d: ", target);  
  
    for (i = 0; i < size; i++)  
        if (arr[i] == target)  
        {  
            count++;  
            printf(" %d ", i+1);  
        }  
  
    if (count == 0)  
        printf("Not found");  
    else  
        printf("\nTotal occurrences: %d", count);  
}
```


Output:

```
Enter the size of the array: 5
Enter 5 elements: 10 22 31 22 47
Enter the search element: 22
Position(s) of 22: 2 4
Total occurrences: 2
```

```
Enter the size of the array: 5
Enter 5 elements: 10 22 31 22 47
Enter the search element: 11
Position(s) of 11: Not found
```

EX 01:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include <stdio.h>

void main(){

    int n, i,arr[20];

    int max,min,max_pos=0,min_pos=0;

    clrscr();

    printf("Enter the number of elements in the array: ");

    scanf("%d", &n);

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++){

        printf("Element %d: ", i);

        scanf("%d", &arr[i]);

    }

    max = arr[0];

    min = arr[0];

    for (i = 1; i < n; i++) {

        if (arr[i] > max)

        {

            max = arr[i];

            max_pos = i;

        }

    }
```

```
        if (arr[i] < min)
        {
            min = arr[i];
            min_pos = i;
        }
    }

    printf("\nLargest element is %d at position %d\n", max, max_pos);
    printf("Smallest element is %d at position %d\n", min, min_pos);
}
```

Output:

```
Enter the number of elements in the array: 5
Enter 5 elements:
Element 0: 12
Element 1: 42
Element 2: 81
Element 3: 35
Element 4: 29

Largest element is 81 at position 2
Smallest element is 12 at position 0
```

EX 02:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include <stdio.h>

void main() {

    int n, i, j, temp, arr[25];

    clrscr();

    printf("Enter the number of elements: ");

    scanf("%d", &n);


    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++) {

        printf("Element %d: ", i);

        scanf("%d", &arr[i]);

    }


    for (i = 0; i < n - 1; i++)

        for (j = 0; j < n - i - 1; j++)

            if (arr[j] > arr[j + 1])

                {

                    temp = arr[j];

                    arr[j] = arr[j + 1];

                    arr[j + 1] = temp;
```

```
}
```

```
printf("\nArray in ascending order:\n");
```

```
for (i = 0; i < n; i++)
```

```
    printf(" %d ", arr[i]);
```

```
printf("\n");
```

```
}
```

Output:

```
Enter the number of elements: 5
```

```
Enter 5 elements:
```

```
Element 0: 12
```

```
Element 1: 36
```

```
Element 2: 70
```

```
Element 3: 43
```

```
Element 4: 88
```

```
Array in ascending order:
```

```
12 36 43 70 88
```

EX 03:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include<stdio.h>

int a[10][10],b[10][10];

int i,j,x,y;

void matrixadd();

void matrixmul();

void readmatrix();

void main(void)
{
    int ch;

    clrscr();

    do
    {

        printf("\nMatrix operations");

        printf("\n1.Addition");

        printf("\n2.Multiplication");

        printf("\n3.Exit");

        printf("\nEnter your choice:");

        scanf("%d",&ch);

        if(ch==1)

            matrixadd();
```

```
        else if(ch==2)
            matrixmul();
        else if(ch==3)
            exit(0);
        else
        {
            printf("Invalid Choice, Exiting!...\n");
            getch();
            exit(0);
        }
    }while(ch==1 ||ch==2 || ch==3);
    getch();
}
```

```
void readmatrix()
{
    printf("\nEnter order of the matrices a and b:");
    scanf("%d %d",&x,&y);
    printf("\nEnter matrix a :\n");
    for(i=0;i<x;i++)
        for(j=0;j<y;j++)
            scanf("%d",&a[i][j]);

    printf("\nEnter matrix b:\n");
```

```
        for(i=0;i<x;i++)
            for(j=0;j<y;j++)
                scanf("%d",&b[i][j]);
    }
```

```
void matrixadd()
{
    readmatrix();
    printf("\nMatrix a+b is:\n");
    for(i=0;i<x;i++)
    {
        for(j=0;j<y;j++)
            printf("%d\t",a[i][j]+b[i][j]);
        printf("\n");
    }
}
```

```
void matrixmul()
{
    readmatrix();
    printf("\n matrix a*b is :\n");
    for(i=0;i<x;i++)
    {
```



```

        for(j=0;j<y;j++)

            printf("%d\t",a[i][j]*b[i][j]);

        printf("\n");

    }

}

```

Output:

Matrix operations

1.Addition

2.Multiplication

3.Exit

Enter your choice:1

Enter order of the matrices a and b:2 2

Enter matrix a :

5 2

7 9

Enter matrix b:

2 5

2 8

Matrix a+b is:

7 7

9 17

Matrix operations

1.Addition

2.Multiplication

3.Exit

Enter your choice:2

Enter order of the matrices a and b:2 2

Enter matrix a :

5 2

7 9

Enter matrix b:

2 5

2 8

Matrix a*b is :

10 10

14 72

EX 04:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include<stdio.h>

#include<conio.h>

long factorial(int);

long find_ncr(int,int);

long find_npr(int,int);

int main()
{
    int n,r;

    long ncr,npr;

    clrscr();

    printf("\nEnter the value for n and r: ");

    scanf("%d %d",&n,&r);

    ncr=find_ncr(n,r);

    npr=find_npr(n,r);

    printf("\n%d c %d = %ld\n",n,r,ncr);

    printf("\n%d p %d = %ld\n",n,r,npr);

    getch();

    return 0;
}
```

```
long find_ncr(int a,int b)
{
    return(factorial(a)/(factorial(b)*factorial(a-b)));
}
```

```
long find_npr(int a,int b)
{
    return(factorial(a)/factorial(a-b));
}
```

```
long factorial(int c)
{
    if(c==1||c==0)
        return 1;
    else
        return c*factorial(c-1);
}
```

Output:

Enter the value for n and r: 5 2

5 c 2 = 10

5 p 2 = 20

EX 05:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include<stdio.h>

#include<string.h>

#include<ctype.h>

void main()

{

    char str[100],c;

    int wcount=0,vcount=0,dcount=0,scount=0,cons=0,i,n;

    clrscr();

    printf("\n*****OUTPUT*****");

    printf("\nEnter a text:\n");

    gets(str);

    n=strlen(str);

    for(i=0;i<n;i++){

        if(isalpha(str[i])>0)

        {

            wcount++;

            c=toupper(str[i]);

            if(c=='A' || c=='E' || c=='I' || c=='O' || c=='U')

                vcount++;

            else
```

```

        cons++;
    }
    else if(isdigit(str[i])>0)
        dcount++;
    else if(isspace(str[i])>0)
        scount++;
}

printf("\nThe number of letters in the text is:%d\n",wcount);
printf("\nThe number of vowels in the text is:%d\n",vcount);
printf("\nThe Number of consonants in the text is %d\n",cons);
printf("\nThe number of spacesin the text is:%d\n",scount);
printf("\nThe number of digits in the text is:%d\n",dcount);
printf("\n*****END*****");
*");
    getch();
}

```

Output:

```

*****OUTPUT*****
Enter a text:
Sacred Heart College 574224

The number of letters in the text is:18
The number of vowels in the text is:7
The Number of consonants in the text is 11
The number of spacesin the text is:3
The number of digits in the text is:6
*****END*****

```

EX 06:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE C PROGRAM TO

Source Code:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str[10][50], *ptr[10], *temp;
    int n, i, j;
    clrscr();
    printf("\nEnter the number of strings MAX 10. \n");
    scanf("%d",&n);
    getchar();
    printf("Enter the strings:\n");
    for (i = 0; i < n; i++){
        gets(str[i]);
        ptr[i] = str[i];
    }

    for (i = 0; i < n-1; i++)
        for (j = i + 1; j < n; j++){
            if (strcmp(ptr[i], ptr[j]) > 0){
```

```

        temp = ptr[i];
        ptr[i] = ptr[j];
        ptr[j] = temp;
    }
}

printf("\nSorted strings in ascending order:\n");
for (i = 0; i < n; i++)
    puts(ptr[i]);

return 0;
}

```

Output:

```

Enter the number of strings MAX 10.
5
Enter the strings:
Date
Cherry
Grape
Apple
Banana

Sorted strings in ascending order:
Apple
Banana
Cherry
Date
Grape

```

EX 07:

DATE:

NAME: ASHWITH FRANK

CLASS: I BCA

PURPOSE: WRITE A C PROGRAM THAT PROMPTS THE USER TO ENTER A STUDENT'S NAME, REGISTRANTION NUMBER, AND MARKS IN THREE SUBJECTS. THE PROGRAM SHOULD THEN CALCULATE AND DISPLAY THE STUDENTS'S TOTAL MARKS, AVERAGE MARKS, AND FINAL GRADE.

Source Code:

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
char getgrade(int);
```

```
void main(void)
```

```
{
```

```
    struct
```

```
    {
```

```
        char name[20];
```

```
        char regno[15];
```

```
        int mark[3];
```

```
    } student;
```

```
    char grade;
```

```
    int total;
```

```
    float average;
```

```
    clrscr();
```

```
    printf("Enter the register number: ");
```



```
scanf("%s", student.regno);

printf("Enter the name of the student: ");
scanf("%s", student.name);

printf("Enter the marks in FIT, COA, Programming in C (out of 100):\n");
scanf("%d %d %d", &student.mark[0], &student.mark[1],
&student.mark[2]);

total = student.mark[0] + student.mark[1] + student.mark[2];
average = total / 3.0;

if (student.mark[0] >= 35 && student.mark[1] >= 35 && student.mark[2]
>= 35)
    grade = getgrade(average);
else
    grade='F';

printf("\n===== STUDENT DETAILS =====\n");
printf("Name      : %s\n", student.name);
printf("Reg. No   : %s\n", student.regno);
printf("Marks     : %d, %d, %d\n", student.mark[0], student.mark[1],
student.mark[2]);
printf("Total      : %d\n", total);
printf("Average   : %.2f\n", average);
```

```

printf("Grade    : %c\n", grade);

printf("=====\n");

getch(); // wait for key press before exit
}

```

```

char getgrade(int avg)
{
    char r;

    if (avg >= 80)
        r = 'A';

    else if (avg >= 60)
        r = 'B';

    else
        r = 'C';

    return r;
}

```

Output:

```

Enter the register number: U05SH25S0002
Enter the name of the student: Ashwith
Enter the marks in FIT, COA, Programming in C (out of 100):
90 95 82

```

```

===== STUDENT DETAILS =====
Name       : Ashwith
Reg. No    : U05SH25S0002
Marks      : 90, 95, 82
Total      : 267
Average    : 89.00
Grade      : A
=====

```